

Annotated Bibliography — Assessment #2

Robin Kim · ISM I · Baseball Analytics Research

Reference 1

Caporale, T., & Collier, T. C. (2013). Scouts versus stats: The impact of Moneyball on the Major League Baseball draft. *Applied Economics*, 45(15), 1983–1990.
<https://doi.org/10.1080/00036846.2011.641933>

This paper tests the Moneyball-era claim that college players are safer MLB draft picks than high schoolers, finding that college draftees performed no better in the majors than high school draftees once selection effects were controlled for. The authors are economics professors at the University of Dayton publishing in a peer-reviewed economics journal, giving the paper strong academic authority and an appropriately quantitative, non-baseball-insider perspective. Its scope is broad and well-suited to this claim, drawing on multiple MLB draft classes from 1965 to 2010, though it examines only the professional draft rather than any HS-level decision directly. I will use this to question whether “more statistical data” inherently means a better decision, which supports the skepticism behind my Hans Rosling framing.

Reference 2

McCue, M., Baker, J., Lemez, S., & Wattie, N. (2019). Getting to first base: Developmental trajectories of Major League Baseball players. *Frontiers in Psychology*, 10, Article 2563.
<https://doi.org/10.3389/fpsyg.2019.02563>

This study traces the pre-MLB pathways of 2,291 American-born MLB players, coding whether each was drafted from a high school, two-year, or four-year program, and finds that 78% of eventual MLB players were not drafted or signed directly out of high school. The authors are sport science researchers at Ontario Tech University, York University, and Cal Poly Pomona, publishing in a peer-reviewed, fully open-access journal, supporting both authority and transparency. Its scope is appropriately broad, covering two decades of debutants league-wide, though it relies on publicly available career statistics rather than original performance data. I will use this to support the point that HS-level statistics alone are a weak stand-alone predictor of MLB readiness.

Reference 3

Efron, B., & Morris, C. (1977). Stein's paradox in statistics. *Scientific American*, 236(5), 119–127.

This classic paper introduces the James–Stein shrinkage estimator, using real 1970 MLB batting averages as its worked example to show that small-sample statistics are unreliable unless “shrunk” toward the group average. The authors, Bradley Efron of Stanford and Carl Morris of Harvard, are among the most influential statisticians of the 20th century, giving the source outstanding authority, and the piece is a purely mathematical, non-partisan demonstration with no stake in any baseball question. Its scope is narrow by design, a short expository magazine piece rather than a full study, so it explains a concept rather than testing a specific HS-baseball hypothesis. I will use this to explain mathematically why a single HS season's stats are noisier than MLB's, so raw comparisons between

the two need caution.